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Lab 3: Component Modelling & Architectural Pattern Selection

Architecture Selection: We chose Layered Architecture for the Personal Subscription Auditor System.

1. Architectural Style Analysis

Layered	Selected. Clear Presentation / Business / Data separation; simple and easy to test. Suits a single-user app with modest scale. Con: some call overhead between layers.
Microservices	Rejected. Independent scaling and deployment are not needed for one user's data; network hops and operational complexity would risk the 2-second dashboard load (NFR-002).
Client-Server	Rejected as the main style. A central server would hold every user's financial data (single point of failure and bigger security exposure), and it still needs internal layering.

2. Written Justification

Architectural Choice	Layered Architecture with three layers: Presentation (Dashboard UI Component), Business (Transaction Import, Subscription Manager, Cancellation Guide, Notification and Audit Report Components) and Data (Database Component).
Reason 1	Separation of concerns maps directly to the requirements: the UI shows the burn rate and renewals (FR-004), business components identify subscriptions (FR-001), send alerts (FR-002), give cancellation guides (FR-003) and handle "Not a Subscription" exclusion (FR-005). A component such as the guide source can be replaced without touching other layers.
Reason 2	The workflow is sequential (import, identify, view burn rate, cancel, export), so requests flow naturally top-down through the layers. It is a small application for an individual user, so one deployable unit with well-defined interfaces (IImportTransactions, IDetectSubscriptions, IManageSubscriptions, ICancellationGuide, INotifyUser, IExportReport, IDataAccess) is simpler than distributed services.
Security Advantage	The UI never touches the database directly; all access goes through the business layer via IDataAccess, so validation is enforced in one place. Only the Transaction Import Component handles bank files, and it parses CSV/OFX data without storing raw banking credentials (NFR-001). The Finance Auditor only receives the exported report through IExportReport.
Performance Benefit	Layers communicate through in-process calls within one application, avoiding the network latency of microservices. The burn rate and upcoming renewals can be read with a single query through IDataAccess, which helps the dashboard load in under 2 seconds (NFR-002).
Diagram	Refer to Lab3_Component_Diagram.png (7 components, 7 interfaces, provided/required notation with <<use>> dependencies).